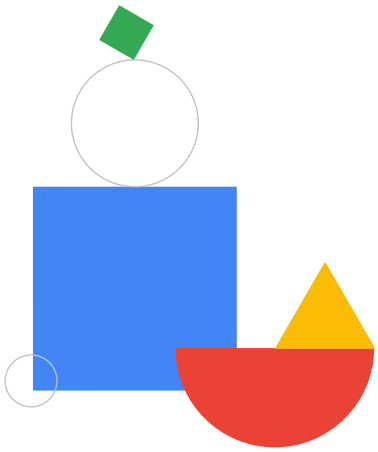
Google Cloud

Installation and Platform Operation



In this module, you will learn the hybrid installation process and complete a lab to install Apigee hybrid on Google Kubernetes Engine.

You will also learn about the tools to manage the hybrid runtime components and configure your cluster resources.

We will also discuss the process to backup and restore your hybrid runtime data.

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Agenda

01 Installation Process

02 Lab: Installing and Managing Apigee Hybrid

03 Platform Operation

04 Backup and Restore

05 Quiz

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Let's start with the installation process. This process guides you through the high-level steps needed to install the Apigee hybrid runtime plane.

In this lecture, we will review these steps and the contents of the hybrid software distribution package.

We will also discuss the strategy used to configure the hybrid runtime plane.

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Process overview

To install Apigee hybrid, you will generally follow the steps below:

- Set up your Google Cloud account, project, and organization.
- Enable APIs and create an Apigee organization, environment group, and environment.
- Calculate the approximate size of the Kubernetes clusters, and create and configure them.
- Download Apigee Helm charts.
- Install the cert-manager required by Apigee hybrid.
- For each cluster, create an overrides.yaml file with your hybrid-specific configuration.
- Apply the configuration.
- Expose the Apigee ingress gateway.
- Create a simple API proxy to verify operation.

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The process to install Apigee hybrid is as follows:

- Set up your Google cloud account, project, and organization.
- Enable relevant APIs and create an Apigee organization, environment group, and environment.
- Determine the size of your Kubernetes clusters based on your API proxy workload and performance requirements,and then create the clusters.
- Download Apigee Helm charts.
- Install the cert-manager required by Apigee hybrid.
- The hybrid software includes configuration for most resources deployed in a cluster. You use an overrides file to override some of the configuration and provide any required properties.
- The configuration specified in this file is then applied to your hybrid clusters.
- Expose the Apigee ingress gateway.
- When all the hybrid services are running, deploy a simple API proxy to verify the installation.

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Helm charts for Apigee hybrid components

Scope	Components	Helm chart
Apigee operator	Apigee operator	`apigee-operator`
Storage	Cassandra	`apigee-datastore`
In-memory storage	Redis	`apigee-redis`
Reporting	Logger Metrics	`apigee-telemetry`
Ingress	Apigee Ingress gateway	`apigee-ingress-manager`
Organization	Apigee Connect Agent MART UDCA Watcher	`apigee-org`
Environment	Runtime Synchronizer	`apigee-env`
Environment group	virtualhost	`apigee-virtualhost`

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The hybrid components are installed and managed by specific Helm charts.

The table is arranged in the order you typically install them in during a fresh installation.

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Setup

The recommended setup tasks to install Apigee hybrid:

1. Create and configure the Kubernetes cluster.

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Follow these steps to install the runtime plane:

First, based on your sizing requirements, create and configure the cluster on your Kubernetes platform.

Configure persistent solid state disk (SSD) storage for Cassandra, the datastore used by Apigee hybrid in the runtime plane.

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Setup

The recommended setup tasks to install Apigee hybrid:

1. Create and configure the Kubernetes cluster.
2. Install cert-manager in the cluster.

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Apigee hybrid uses cert-manager to manage certificates used by the various components in the runtime plane.

Cert-manager is a native Kubernetes certificate management controller that helps with issuing and renewing of certificates.

You install cert-manager from a GitHub repository using the kubectl apply command.

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Setup

The recommended setup tasks to install Apigee hybrid:

1. Create and configure the Kubernetes cluster.
2. Install cert-manager in the cluster.
3. Create a directory structure for certificates, service accounts, and override files.

```
$mkdir ${HOME}/apigee/helm
```

Google Cloud

Next, set up your administration machine with a recommended directory structure on the file system.

Create a subdirectory named helm under the HOME/apigee directory. You can give the directory any name you want.

This directory is used to store helm charts and other files for TLS certificates, service accounts, override configuration, and hybrid tools, which are all used in the hybrid runtime installation process.

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Setup

The recommended setup tasks to install Apigee hybrid:

1. Create and configure the Kubernetes cluster.
2. Install cert-manager in the cluster.
3. Create a directory structure for certificates, service accounts, and override files.
   $mkdir ${HOME}/apigee/helm
4. Download the Apigee hybrid Helm charts.

Google Cloud

Next, download the Apigee hybrid Helm charts from Artifact Registry. Apigee hybrid uses Helm charts to install and manage the runtime plane components.

Helm is an open-source tool that helps manage and deploy applications in a Kubernetes cluster using a packaging format called a Helm chart.

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Setup

The recommended setup tasks to install Apigee hybrid:

1. Create and configure the Kubernetes cluster.
2. Install cert-manager in the cluster.
3. Create a directory structure for certificates, service accounts, and override files.
   $mkdir ${HOME}/apigee/helm
4. Download the Apigee hybrid Helm charts.
5. Add environment variables to your shell.

Google Cloud

Set up shell environment variables for your Helm chart directory.

These variables are used for cluster authentication and to run other hybrid installation commands.

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Setup

The recommended setup tasks to install Apigee hybrid:

6. Create the necessary certificates, keys, and service accounts.

Google Cloud

A TLS certificate and key are needed for the runtime ingress gateway that handles API proxy traffic. For trial installations of Apigee hybrid, you can use a self-signed certificate.

Components in the hybrid runtime plane use Google Cloud service accounts to communicate with the management plane and with other Google Cloud services.

You can create these service accounts by using the create-service-account tool that is included in the Apigee hybrid distribution package.

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Setup

The recommended setup tasks to install Apigee hybrid:

6. Create the necessary certificates, keys, and service accounts.
7. Create the apigee namespace.

Google Cloud

Create the namespace in the Kubernetes cluster that is used for the Apigee hybrid runtime plane resources.

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Setup

The recommended setup tasks to install Apigee hybrid:

6. Create the necessary certificates, keys, and service accounts.
7. Create the apigee namespace.
8. Create an overrides file to configure the cluster.

Google Cloud

Apigee hybrid uses defaults for most cluster configuration settings for the hybrid runtime plane components.

However, a few settings do not have defaults, and you can override others.

You provide values for these settings in an overrides.yaml configuration file.

The overrides.yaml file is usually created in the Helm charts home directory.

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Setup

The recommended setup tasks to install Apigee hybrid:

6. Create the necessary certificates, keys, and service accounts.
7. Create the apigee namespace.
8. Create an overrides file to configure the cluster.
9. Enable synchronizer to authenticate against the management plane.

Google Cloud

After the overrides.yaml file is created, you enable the required permissions for the synchronizer runtime component to authenticate against the hybrid management plane.

Also, enable the runtime plane components to publish analytics and debugging information to the control or management plane.

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Setup

The recommended setup tasks to install Apigee hybrid:

6. Create the necessary certificates, keys, and service accounts.
7. Create the apigee namespace.
8. Create an overrides file to configure the cluster.
9. Enable synchronizer to authenticate against the management plane.
10. Install the Apigee hybrid runtime.

Google Cloud

You first install Apigee custom resource definitions (CRDs) to create and manage custom resources in your cluster.

Make sure to have the latest version of Helm installed on your machine that is used to run the *helm* commands.

After the CRDs are installed, you use the Apigee hybrid Helm charts to install the runtime components one at a time in sequence in your cluster.

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Setup

The recommended setup tasks to install Apigee hybrid:

6. Create the necessary certificates, keys, and service accounts.
7. Create the apigee namespace.
8. Create an overrides file to configure the cluster.
9. Enable synchronizer to authenticate against the management plane.
10. Install the Apigee hybrid runtime.
11. Expose the Apigee ingress gateway.

Google Cloud

Finally, set up the Kubernetes service for your Apigee ingress gateway. The Kubernetes service is needed for each ingress deployment to obtain an IP address that can be exposed. Client calls to proxies will invoke a hostname that will resolve to this IP address.

There are two options for providing a Kubernetes service to assign the IP address. The recommended option for production environments is to create a custom Kubernetes service for each Apigee ingress gateway. For non-production environments or to test initial traffic through the Apigee ingress gateway, Apigee hybrid provides a default Kubernetes service for each ingress deployment.

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Overrides strategy

```
gcp:
  region: gcp-region
  projectID: gcp-project-id

k8sCluster:
  name: cluster-name
  region: cluster-region

org: org-name

instanceID: “unique-id”

virtualhosts:
- name: env-group-name
  selector:
    app: apigee-ingressgateway
    ingress_name: INGRESS_NAME
  sslCertPath: ./certs/cert-name.pem
  sslKeyPath: ./certs/key-name.key

envs:
- name: environment-name
  serviceAccountPaths:
    synchronizer: ./sa/sync-sa.json
    udca: ./sa/udca-sa.json
    runtime: ./sa/runtime-sa.json
...
```

Sample overrides.yaml file fragment

The Apigee hybrid installation process uses an overrides strategy:

- Default values are provided in the values.yaml file, which should never be directly edited.
- The default values can be overridden by configurations supplied by you in a yaml file called "overrides.yaml."
- There can be more than one "overrides" file. It is a best practice to version your override files in source control.
- Name your override files to convey the use case for which they are created.

For example: "*official-prod.yaml*" or "*experimental-dev.yaml*"

- Overrides may describe one or more components of a configuration or the entire configuration.

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When you install Apigee hybrid, you provide configuration properties that are used to configure hybrid components in your cluster.

The properties override the default values that come with the hybrid software distribution.

Using this strategy, you can have multiple override files that configure your cluster with different resource configurations, based on certain criteria. For example, you may have different config settings for a production cluster versus a non-production cluster.

You can also have separate override files where each file configures all the components needed for a single runtime environment in the cluster.

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Override file versioning

- Over time, you will accumulate multiple versions of the apigee tools and changes to override files.
- It is a best practice to maintain versions of the tool and overrides files in source control.
- If rollbacks become necessary, you can apply the previous version of the files to your cluster.
- It is recommended that you test and practice your installation, upgrade, and rollback procedures before production installation.

Google Cloud

It is a best practice to maintain your override.yaml files in source control.

This allows versioning of your configuration and makes it easier to roll back to a previous version if needed.

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Namespaces

Kubernetes Namespaces used by Apigee Hybrid

Namespace	Purpose
apigee	Contains runtime plane components like runtime, cassandra, and UCDA.
apigee-system	Used to deploy and maintain ApigeeDeployment (AD) resources and validate deployment configurations.
cert-manager	Provides PKI certificate management for runtime plane components.
default	Always present in Kubernetes; the namespace given to services if one isn't supplied.

Google Cloud

The Apigee hybrid installation process creates multiple namespaces in your Kubernetes cluster.

These namespaces logically group the runtime components by the type of functionality they provide.

The apigee-system, and cert-manager namespaces contain system components needed for creating the hybrid workload components including the Apigee ingress gateway, and for certificate management.

All of the hybrid workload components are deployed in the apigee namespace in your cluster.

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Verifying the installation

To verify the installation, create a basic API proxy that tests the connectivity between the management plane, runtime plane components, and backend.

If you don't have your own test backend, consider *httpbin.org*.

The basic API proxy should cover the following use cases:

- API key verification
- Backend connectivity
- Quota and/or spike arrest

Verify the following in the management plane via the hybrid UI:

- Performance tab
- Trace
- Analytics

You can also verify the following in the Google Cloud console:

- Log entries in Cloud Logging
- Metrics in Cloud Monitoring

Google Cloud

After Apigee hybrid is installed, you can verify that it is operating as expected.

Create a test environment in the hybrid UI and configure this environment in the hybrid runtime plane.

Then, create or upload a simple API proxy using the hybrid UI and deploy it to the test environment. This test API proxy can use a mock backend as the target of the API.

Send multiple requests to the API. You can use curl, Postman, or any REST client to generate HTTP requests.

Confirm that the API proxy is returning the expected response from the backend.

You can add API key verification or quota policies to the test API proxy to verify that the runtime datastore is functioning as expected. This will confirm that the MART, Message Processor, and Cassandra runtime components are functioning as expected.

In addition, you can start Trace sessions to debug your proxy and view Analytics in the hybrid UI.

Using the Google Cloud console, you can view log messages to troubleshoot the hybrid runtime components and view metrics to track resource usage of the cluster components.

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Google Cloud

We will now do a lab: Installing and Managing Apigee Hybrid.

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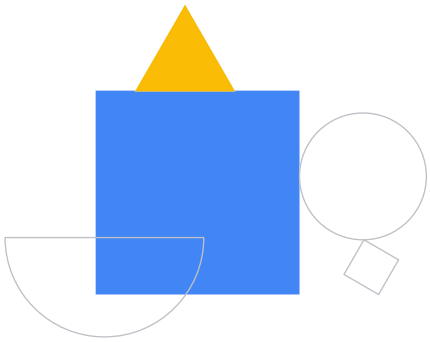
Lab Intro

Installing and Managing
Apigee Hybrid



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In this lab, you install and configure the Apigee hybrid runtime plane in a Kubernetes cluster on GKE. You then create Apigee hybrid environments and an environment group using the Apigee hybrid UI.

Next, you deploy simple API proxies to the Apigee hybrid environments and validate the environments by testing the API proxies.

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Google Cloud

In this lecture we discuss the tools used to operate the Apigee hybrid runtime platform.

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Runtime component management: helm upgrade

To make a configuration change to a hybrid runtime plane component, edit your overrides file and apply the changes with helm.

For example, to change the replica count on the runtime MP component:

- Edit the overrides.yaml file to update the replicaCountMin/replicaCountMax entries.
- Use helm to apply the change to the cluster and upgrade the env:

```
$ helm upgrade {env_release_name} apigee-env/
--install --namespace {apigee-namespace} --set
env={env_name} --atomic -f overrides.yaml
```

overrides.yaml

```
...
runtime:
  nodeSelector:
    ...
  replicaCountMin: 2
  replicaCountMax: 4
...
```

Google Cloud

Use the helm upgrade command to apply configuration changes in your overrides.yaml file to the cluster.

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Runtime component management: kubectl

- kubectl commands can be used to fetch information about the status of various objects in the cluster:
  - To get a list of all pods in the *apigee* namespace:

    ```
    $ kubectl get pods -n apigee
    ```

  - For more detailed information about a given pod:

    ```
    $ kubectl describe pods {POD_NAME} -n apigee
    ```

    where {POD_NAME} is obtained from the get pods command above.

  - To get a shell for an apigee-runtime pod:

    ```
    $ kubectl exec -it -n apigee {runtime_pod_name} --container apigee-runtime
    -- /bin/bash
    ```

- Details on kubectl and the list of commands are on the Kubernetes docs page.

Google Cloud

Kubectl enables you to manage and control Kubernetes clusters.

You can use kubectl to list and describe cluster resources, like pods, and check on their current status.

You can also start command shells to access the containers running in those pods for debugging purposes.

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ApigeeDeployment

- Apigee hybrid uses a Kubernetes Custom Resource Definition (CRD) called ApigeeDeployment (AD) to define, update, and release stateless Apigee hybrid components in the cluster.
- ApigeeDeployments can be managed using kubectl commands. To get a list of all the ADs in your cluster, run:

```
$kubectl get ad -n {namespace}
NAME                                        AGE
apigee-mart-my_org                          4h
apigee-runtime-my_org-my_env                4h
apigee-synchronizer-my_org-my_env           4h
apigee-udca-my_org-my_env                   4h
```

- Hybrid uses a custom controller called apigee-operators (ao), along with the CRD, to maintain the desired state of hybrid resources. The controller:
  - Validates the ApigeeDeployment configuration using an admission webhook before persisting it in the cluster.
  - Implements the release of message processors in the cluster.

Google Cloud

The Kubernertes CustomResourceDefinition API resource allows you to define custom resources in your cluster.

Using a CRD object, you can create a custom resource with a name and schema that you specify.

Apigee hybrid uses a CustomResourceDefinition called ApigeeDeployment.

The ApigeeDeployment objects can be managed by using kubectl commands.

Along with the Apigee Operators custom controller, they validate deployment configurations and deploy hybrid runtime resources in the cluster.

The **apigee-deployment-admissionhook** is a HTTP callback used for validating requests for changes to ApigeeDeployment resources.

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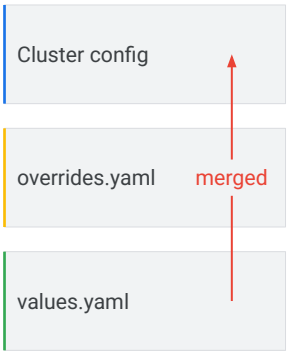
Configuring hybrid

- Apigee hybrid maintains its default component configuration in the file:

  `values.yaml`
- As part of the installation process, you create an overrides.yaml file that typically includes only a subset of the configuration properties in values.yaml.
- When you apply a configuration to a cluster, your overrides are merged with the defaults to create the complete Kubernetes cluster configuration.
- Not all properties in the values.yaml file are editable, and some overrides are mandatory, such as *gcpProjectID*.
- If you want to modify a component's defaults, first copy its YAML from *values.yaml* into your overrides file, and modify it there. DO NOT modify the values.yaml file directly.



Google Cloud

Apigee hybrid uses a layered approach to configure your cluster.

The default configuration is stored in a file called values.yaml in the Apigee hybrid helm installation directories of the hybrid distribution.

To install Apigee hybrid, you create an overrides configuration file that includes mandatory properties and other properties that override the default values in the values.yaml file.

Not all default properties can be overridden. For those that can and should be modified, the best practice is to provide the new values in the overrides.yaml config file instead of directly modifying the values.yaml file. This approach preserves your modified values during product upgrades.

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Using overrides

- You can create as many overrides files as you want, where each one serves a specific requirement.
  - For example, you might have an overrides file that tunes your cluster for production, and another for creating a testing cluster. You can then maintain these files in your source control system.
- Use the -f option with with the helm upgrade or helm install command to specify the location and name of an overrides file:

```
$helm upgrade test-1-env apigee-env/ \
--namespace apigee --atomic \
--set env=test-1-env \
-f test-1-env-overrides.yaml
```

- It is a good practice to copy and modify the overrides file that most closely matches your installation requirements.
- See the full [configuration properties] documentation for reference.

Google Cloud

It is also a best practice to maintain multiple override files in source control.

You can have one overrides file for your production clusters and a separate one for a non-production cluster.

You can also have separate override files for each hybrid environment in the cluster.

A good practice is to maintain a set of override files where each file contains only the configuration required for each group of components.

For example, you can have a common.yaml overrides file for components like cassandra, logging, and metrics.

Another file can contain the configuration properties for the hybrid organization and resources like mart and apigee connect.

Environment-level properties for the runtime, synchronizer, and udca components can be in their own overrides file, one file per hybrid environment.

Virtual host configuration can also be in its own overrides configuration file.

Using this strategy, you can control changes to a subset of components in the cluster by updating and applying only the specific overrides file.

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Checking for configuration errors

Check for errors in your hybrid cluster configuration:

- Use the –dry-run flag to check for errors in your overrides.yaml file without modifying your cluster.

```
$ kubectl apply -k  apigee-operator/etc/crds/default/
$ helm upgrade operator apigee-operator/ --install --namespace apigee \
--atomic -f overrrides_file.yaml --dry-run=server
```

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You can use the dry-run flag to check for errors in your configuration without making any changes to the cluster.

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Automated issue surfacing

Automated issue surfacing (AIS) provides quick information about ongoing issues detected within your Hybrid cluster.

- Runtime watcher scans to determine if there are any configuration issues.
- AIS links to documentation to help with troubleshooting and resolution.
- AIS looks for known, common, system detectable issues.
- AIS cannot detect all the issues within a cluster.
- You can check for any existing issues with the kubectl get **apigeeissues** command.

```
$ kubectl -n apigee get apigeeissues
```

Google Cloud

Automated issue surfacing (AIS) provides quick information about ongoing issues detected within your Hybrid cluster.

The Apigee runtime watcher component automatically scans the control plane and Kubernetes API server state to determine if there are any configuration issues. By default, the scanning happens every 60 seconds. You can change the interval or disable scanning if you prefer.

When AIS detects an issue, it creates a new instance of ApigeeIssue within the Kubernetes API server. These instances contain information about the issues and links to documentation on the specific issues.

When you resolve issues, they will automatically be deleted from the Kubernetes API server when the scanning determines they are no longer occurring.

Automated issue surfacing only looks for known, common, system detectable issues and cannot detect all the issues within a cluster.

You can check for any existing issues with the kubectl get apigeeissues command.

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Support for custom annotations

You can create custom annotations for the following properties:

- **cassandra**
- **connectAgent**
- **logger**
- **mart**
- **metrics**
- **runtime**
- **synchronizer**
- **udca**
- **watcher**

overrides.yaml

```
...
runtime:
  annotations:
   businessunit:"bu1"

...
```

Google Cloud

Annotations are key/value maps used to attach metadata to Apigee hybrid Kubernetes pods.

You can create custom annotations for the following properties:

- `cassandra`
- `connectAgent`
- `logger`
- `mart`
- `metrics`
- `runtime`
- `synchronizer`
- `udca`
- `watcher`

To add a custom annotation, add a stanza to the `overrides.yaml` file for the respective component.

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Google Cloud

In this lecture, we discuss procedures to perform backups of your hybrid runtime plane datastore.

We will also discuss how to restore your cluster from a backup.

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Runtime Cassandra

- Apigee hybrid uses a Cassandra ring installed in the runtime plane to provide data persistence for KMS, KVM, Quota, and other data needed by API proxies.
- Cassandra is a replicated database that is configured to have at least 3 copies of your data in each cluster.
- Cassandra uses streaming replication to maintain the data replicas in each region.
- In hybrid, Cassandra backups are not enabled by default. It is a good practice, however, to enable Cassandra backups in case your data is accidentally deleted.

Google Cloud

Apigee hybrid uses a Cassandra database to store your hybrid runtime data.

Cassandra is a replicated database that is configured to have at least 3 copies of your data in each cluster. Cassandra uses streaming replication and read repairs to maintain the data replicas at any given point.

The Cassandra runtime data store is deployed to your Kubernetes cluster as a StatefulSet. The Cassandra pod reads and writes data using a PersistentVolume. If the pod is deleted and re-created, it will continue to have access to the PersistentVolume.

In hybrid, Cassandra backups are **not** enabled by default. It is a good practice, however, to enable Cassandra backups in case your data is accidentally deleted.

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Backup overview

There are **two** methods to configure backups:

1. **Using Apigee hybrid CSI backup and restore,** which uses Kubernetes CSI (Container Storage Interface) snapshots.
2. **Using non-CSI hybrid backup and restore**, which copies schema and other data to backup files.

For both methods, you'll configure backup in your overrides.yaml file.

Google Cloud

You can use one of two methods to configure backups.

Using Apigee hybrid CSI backup and restore, which uses Kubernetes CSI (Container Storage Interface) snapshots. This method is recommended for hybrid instances hosted in Google Cloud, AWS, or Azure.

Using non-CSI hybrid backup and restore, which copies schema and other data to backup files. This is the recommended method for on-prem installations.

For both methods, you'll configure backup in your overrides.yaml file. Apigee recommends you make a copy of the overrides.yaml file, so that you can reuse it during the recovery process.

Let’s take a look at using the Apigee hybrid CSI backup and restore method.

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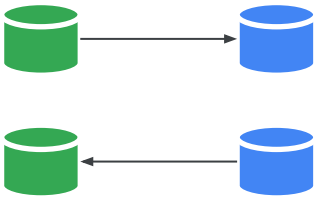
Cassandra CSI backup and restore

Starting with Apigee hybrid 1.9, you can backup and restore your hybrid data using CSI (Container Storage Interface) snapshots.

- CSI backup generates disk snapshots and stores them as encrypted data in Cloud Storage.
- CSI backup does not need a Cloud Storage bucket or a remote server to store backup data.
- CSI backup is recommended for hybrid instances hosted in Google Cloud, AWS, or Azure.

For more information, see Cassandra CSI backup and restore.



Google Cloud

Starting with Apigee hybrid 1.9, you can backup and restore your hybrid data using CSI (Container Storage Interface) snapshots. CSI backup generates disk snapshots and stores them as encrypted data in Cloud Storage. CSI backup does not need a Cloud Storage bucket or a remote server to store backup data.

CSI backup is recommended for hybrid instances hosted in Google Cloud, AWS, or Azure.

For more information, see Cassandra CSI backup and restore.

Now let’s take a look at how to schedule backups in Cloud Storage and in a remote server, using non-csi backup and restore methods.

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Cassandra non-CSI backups

Entities that are backed up:

- Cassandra schema including the user schema (Apigee keyspace definitions)
- Cassandra partition token information per node
- A daily snapshot of the Cassandra data

Backup data storage:

- Backed-up data is stored in a Cloud Storage bucket that you must create.
- You can also store backups on a server file system.

Google Cloud

Cassandra non-CSI backups are best suited to schedule backups in Cloud Storage and in a remote server.

The entities that are backed up in this process include the Cassandra schema, partition token information per node, and a daily snapshot of the data.

The backup is stored in a Cloud Storage bucket that you must create before enabling the Cassandra backup.

You can also back up your datastore to a server file system. This topic is covered in later slides in this section.

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Scheduling backups

Backups are scheduled as cron jobs in the runtime plane. To schedule Cassandra backups:

1. Create a service account with the standard roles/storage.objectAdmin role. This role allows you to write backup data to Cloud Storage.

   Use the `create-service-account` tool that is in the $APIGEE_HELM_CHARTS_HOME/apigee-operator/etc directory.

   ```
   $ ./tools/create-service-account --env prod \
     --profile apigee-cassandra --dir $APIGEE_HELM_CHARTS_HOME/apigee-datastore
   ```

Google Cloud

To schedule backups, you must perform a set of steps.

First, create a service account with the storage.objectAdmin role. This service account role allows you to back up data to Cloud Storage.

To create this service account, use the create-service-account tool.

Store the service account json key file in the apigee-datastore Helm chart directory.

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Scheduling backups

Backups are scheduled as cron jobs in the runtime plane. To schedule Cassandra backups:

1. Create a service account with the standard roles/storage.objectAdmin role. This role allows you to write backup data to Cloud Storage.

   Use the `create-service-account` tool that is in the $APIGEE_HELM_CHARTS_HOME/apigee-operator/etc directory.

   ```
   $ ./tools/create-service-account --env prod \
     --profile apigee-cassandra --dir $APIGEE_HELM_CHARTS_HOME/apigee-datastore
   ```

2. Create a Cloud Storage bucket. Specify a reasonable data retention policy (15 days) for the bucket.

Google Cloud

Create a storage bucket in Cloud Storage.

Specify a data retention policy for the bucket. 15 days is a reasonable period.

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Scheduling backups

Backups are scheduled as cron jobs in the runtime plane. To schedule Cassandra backups:

1. Create a service account with the standard roles/storage.objectAdmin role. This role allows you to write backup data to Cloud Storage.

   Use the `create-service-account` tool that is in the $APIGEE_HELM_CHARTS_HOME/apigee-operator/etc directory.

   ```
   $ ./tools/create-service-account --env prod \
     --profile apigee-cassandra --dir $APIGEE_HELM_CHARTS_HOME/apigee-datastore
   ```

2. Create a Cloud Storage bucket. Specify a reasonable data retention policy (15 days) for the bucket.
3. Edit the overrides.yaml file to configure Cassandra backup:

overrides.yaml

```
cassandra:
  ...
  backup:
    enabled: true
    serviceAccountPath: "apigee-cassandra.json"
    dbStorageBucket: "cassandra-bkup"
    schedule: "45 23 * * 6"
    cloudProvider: "GCP"
 ...
```

Google Cloud

Update your overrides.yaml file and add the backup configuration properties to the cassandra stanza.

These properties enable the backup, set the path to the service account file and the Cloud Storage bucket, and configure a cron schedule to run the backup.

For example, the configuration shown will schedule a backup using the service account json file generated from the create-service-account command above.

The data is backed up to the Cloud Storage bucket at gs://cassandra-bkup. The cron schedule to run the backup is as configured.

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Scheduling backups

Backups are scheduled as cron jobs in the runtime plane. To schedule Cassandra backups:

1. Create a service account with the standard roles/storage.objectAdmin role. This role allows you to write backup data to Cloud Storage.

   Use the `create-service-account` tool that is in the $APIGEE_HELM_CHARTS_HOME/apigee-operator/etc directory.

   ```
   $ ./tools/create-service-account --env prod \
     --profile apigee-cassandra --dir $APIGEE_HELM_CHARTS_HOME/apigee-datastore
   ```

2. Create a Cloud Storage bucket. Specify a reasonable data retention policy (15 days) for the bucket.
3. Edit the overrides.yaml file to configure Cassandra backup:
4. Apply the changes to the runtime cluster:

   ```
   $ helm upgrade datastore apigee-datastore/ --namespace apigee \
   --atomic -f overrides.yaml
   ```

   When you apply the backup configuration, Kubernetes recreates the Cassandra nodes, so allow for enough time before backup starts.

overrides.yaml

```yaml
cassandra:
  ...
  backup:
    enabled: true
    serviceAccountPath: "apigee-cassandra.json"
    dbStorageBucket: "cassandra-bkup"
    schedule: "45 23 * * 6"
    cloudProvider: "GCP"
 ...
```

Google Cloud

Use the *helm upgrade* command to apply the override configuration changes to the cluster.

Allow for sufficient time in the cron schedule for the backup to begin after the configuration changes are applied to the cluster.

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Verify the backup job

5. Run the following command to verify the backup job:

```
$kubectl get cronjob -n apigee
```

```
NAME                      SCHEDULE     SUSPEND   ACTIVE   LAST SCHEDULE   AGE
apigee-cassandra-backup   33 * * * *   False     0        <none>          94s
```

Google Cloud

To verify the backup job, run the following command: kubectl get cronjob -n apigee.

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Viewing backup logs

- When a backup job is scheduled and running, you can view progress by monitoring the backup pod log.
- Get a list of the pods to determine the backup pod name:
  `$kubectl get pods -n <namespace>`

```
NAME                                        READY     STATUS      RESTARTS   AGE
apigee-cassandra-backup-1554515580-pff6s    0/1       Running     0          54s
```

- Run the kubectl logs command to view the pod log. Partial output follows:
  `$kubectl logs -f apigee-cassandra-backup-1554515580-pff6s`

```
...
INFO: Backup 20190406190808 completed
waiting result
to get schema from 10.32.0.28
INFO: /tmp/schema.cql has been generated
...
```

```
...
Copying from <TDIN>...
/ [1 files][    0.0 B/    0.0 B]
Operation completed over 1 objects.
INFO: backup created tarball and
transferred the file to
gs://cassandra-bkup/apigeecluster/dc-1
finished uploading schema.cql
```

Google Cloud

The backup job creates a schema.cql file and uploads it to the Cloud Storage bucket. It then instructs the Cassandra node to back up the data and waits for the data to be uploaded.

You can monitor a backup job that has started based on your configured cron schedule.

First, use the kubectl get pods command to determine the name of the pod running the backup job.

Run the kubectl logs command, passing in the name of the pod running the backup job.

You will see output in the pod log as the backup job executes.

Note the backup snapshot timestamp. It is used to configure your override properties when your backup is restored.

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Restoring Cassandra in a single region

In a single region deployment, Apigee hybrid is deployed in a single data center or a region.

The Cassandra backup can reside either on Cloud Storage or on a remote server.

To restore Cassandra backups:

1. Confirm that the Kubernetes cluster you are restoring to does not have a prior Apigee hybrid installation. Verify that there are no pods in the Apigee namespace:

```
$kubectl get pods -n apigee
```

2. Otherwise, follow steps to decommission a hybrid region using helm by deleting the Apigee hybrid components one at a time:

```
$helm -n apigee delete {component_release_name}
```

Google Cloud

In a single region deployment, Apigee hybrid is deployed in a single data center or a region. If you have multiple Apigee organizations in your deployment, the restore process restores data for all the organizations. In a multi-organization setup, you cannot restore a specific organization.

In your configuration, the Cassandra backup can reside either on Cloud Storage or on a remote server.

Confirm that the Kubernetes cluster you are restoring to does not have a prior Apigee hybrid installation. If you are restoring to the existing cluster, use the commands shown to decommission the existing Apigee hybrid installation.

For more details on how to decommission an existing region, view the documentation.

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Restoring Cassandra in a single region

3. Update the overrides.yaml file with the restore details::
   a. Specify the namespace element.
   b. Add the cassandra.restore properties.
      - snapshotTimestamp is the timestamp associated with the backup you are restoring.
   c. Set backup.enabled to false.

overrides.yaml

```
namespace: your-restore-namespace
...
cassandra:
  hostNetwork: false
  ...
  restore:
    enabled: true
    snapshotTimestamp: "20190417002207"
    serviceAccountPath: "apigee-cassandra.json"
    dbStorageBucket: "cassandra-bkup"
    cloudProvider: "GCP"
  ...
  backup:
    enabled: false
...
```

Google Cloud

Modify the overrides yaml file to specify the namespace element.

Add the restore configuration properties to the cassandra stanza.

Specify the timestamp of the backup snapshot to restore, the path to the service account file, and the Cloud Storage bucket.

Disable backup when restoring.

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Restoring Cassandra in a single region

4. Install all the Apigee hybrid runtime components using the helm commands as previously described.

Google Cloud

Install all the Apigee hybrid runtime components using the helm commands as previously described when installing the runtime plane.

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Restoring Cassandra in a single region

5. Verify the restoration job progress and confirm that `apigeeds` and all the other pods are up:

```
$kubectl get apigeeds -n apigee
$kubectl get pods -n apigee
```

Google Cloud

Verify the restoration job progress and confirm that apigeeds and all the other pods are up. Upon successful completion of the restore and confirmation that the runtime components are healthy, it is recommended to configure a backup on the cluster.

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Viewing restore logs

- You can check the restore job logs to search for any errors:

```
$kubectl get pods -n <namespace>
NAME                          READY     STATUS      RESTARTS   AGE
apigee-cassandra-restore-b4lgf 0/1      Completed   0          51m
```

- Run the kubectl logs command to view the pod log. Partial output follows:

```
$kubectl logs -f apigee-cassandra-restore-b4lgf -n <namespace>
```

```
...
INFO: download successfully extracted the
backup files from
gs://cassandra-bkup/apigeecluster/dc-1
   finished downloading schema.cql
   to create schema from 10.32.0.28
...
```

```
...
INFO: Restore 20190405011309 completed
INFO:
./apigee/data/cassandra/data/ks1/user-9fbae960571411e9
9652c7b15b2db6cc restored successfully
INFO: Restore 20190405011309 completed
waiting on waiting nodes $pid to finish  106
Restore finished
```

Google Cloud

To monitor the progress of the restore job, first use the kubectl get pods command to determine the name of the pod running the job.

Run the kubectl logs command, passing in the name of the pod running the restore job.

You will see output in the pod log as the restore job executes and completes.

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Configuring file system backups

You can store backups of your Cassandra database to compressed files in the file system of a server you specify.

Follow these steps to configure file system backups:

1. Set up the server and configure SSH.
   - After setup and configuration, verify that the server can be reached from the apigee-cassandra pods in the runtime plane over SSH.
2. Set up the backup schedule and destination.

Google Cloud

You can store backups of your Cassandra database to compressed files in the file system of a server you specify.

Backups occur on a schedule that you specify in your overrides file. The connection to the server is by secure SSH.

To configure a file system backup, you must first set up a server and install and configure SSH. As part of this process, you create a key pair.

The public key is copied to the server, and the location of the private key is configured in your overrides config file.

Test the SSH connection between your Cassandra pods in the runtime cluster and your backup server.

Next, you configure your overrides.yaml file with the backup configuration.

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Configuring file system backups - overrides.yaml

To set up the backup schedule and destination:

1. Edit your overrides.yaml config file to add entries for the cassandra backup. Specify:
   - Path to the SSH private key file
   - File system server IP address
   - Storage location on the file system to contain the backup cloudProvider property with value "HYBRID"
   - Cron schedule for the backup
2. Apply the changes to your cluster using helm:

```
$ helm upgrade datastore apigee-datastore/ --install \
  --namespace apigee --atomic -f overrides.yaml
```

3. Verify the backup job:

```
kubectl get cronjob -n apigee
```

overrides.yaml

```
cassandra:
  ...
  backup:
    enabled: true
    keyFile: "./keys/ssh_private.key"
    server: "34.56.78.90"
    storageDirectory:
"/home/apigee/cassbackup"
    cloudProvider: "HYBRID"
    schedule: "45 23 * * 6"
 ...
```

Google Cloud

Edit your overrides.yaml configuration file to add the backup properties.

Set the enabled property to true, and provide the path to the SSH private key file and the backup server IP address.

Set the directory location where the backup files are to be stored on the backup server.

Set the required *cloudProvider* property to HYBRID, and set the cron schedule for the backup.

Apply the configuration changes to your cluster using helm.

Finally, verify the backup job.

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Configuring file system restore

- The Cassandra backup can also reside on a remote server.
- The process for restoring is the same as Cloud Storage backups, except the overrides.yaml is configured differently.
- Modify the overrides-restore.yaml file:
  a. Specify the namespace element.
  b. Add the cassandra.restore properties.
     - snapshotTimestamp is the timestamp associated with the backup you are restoring.

overrides.yaml

```
namespace: your-restore-namespace
...
cassandra:
  ...
  restore:
    enabled: true
    keyFile: "./keys/ssh_private.key"
    server: "34.56.78.90"
    storageDirectory:
"/home/apigee/cassbackup"
    cloudProvider: "HYBRID"
    snapshotTimestamp: "20190417002207"
 ...
```

Google Cloud

The Cassandra backup can also reside on a remote server, and the process to restore is the same as mentioned in the previous slides on restoring cassandra in a single region.

For non-cloud storage restores, modify the overrides.yaml file to specify the namespace element.

Add the restore configuration properties to the cassandra stanza, for example, the path to the SSH private key file, the file system server IP address, the storage location on the file system to contain the backup, and the cloudProvider property with value "HYBRID".

Specify the timestamp of the backup snapshot to restore and the other properties used to configure the backup.

Then continue the restore process as you would for restoring Cassandra in a single region.

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Agenda

01 Installation Process

02 Lab: Installing and Managing Apigee Hybrid

03 Platform Operation

04 Backup and Restore

05 Quiz

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Question #1

Question

Which of the following properties cannot be configured in the overrides.yaml file used in the hybrid installation process?

A. Google Cloud project ID

B. Namespace

C. Service account files

D. API proxy definition

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Question #1

Answer

Which of the following properties cannot be configured in the overrides.yaml file used in the hybrid installation process?

A. Google Cloud project ID

B. Namespace

C. Service account files

D. API proxy definition

Google Cloud

Explanation:

A. Incorrect. The Google Cloud project id must be configured in the overrides.yaml file to identify the project for sending log and metrics data.

B. Incorrect. The namespace property identifies the Kubernetes cluster where the Apigee hybrid components are installed. It is optional and defaults to apigee.

C. Incorrect. Service accounts files are required to be configured for most of the hybrid runtime components.

D. Correct! API proxy definitions are created via the hybrid UI or Apigee API in the management plane and downloaded by the synchronizer to the runtime plane.

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Question #2

Question

Which of these commands are used to install the Apigee hybrid cassandra component?

A. helm upgrade datastore apigee-cassandra/ --install

B. helm install datastore apigee-datastore/

C. helm install datastore apigee-cassandra/

D. helm upgrade datastore apigee-datastore/ --install

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Question #2

Answer

Which of these commands are used to install the Apigee hybrid cassandra component?

A. helm upgrade datastore apigee-cassandra/ --install

B. helm install datastore apigee-datastore/ ✓

C. helm install datastore apigee-cassandra/

D. helm upgrade datastore apigee-datastore/ --install ✓

Google Cloud

Explanation:

A. Incorrect. apigee-cassandra is not the correct name for the folder that contains the datastore Helm chart.

B. Correct! This command along with additional arguments is used to install the Apigee hybrid cassandra component.

C. Incorrect. apigee-cassandra is not the correct name for the folder that contains the datastore Helm chart.

D. Correct! This command along with additional arguments is used to install or upgrade the Apigee hybrid cassandra component.

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Question #3

Question

Which of the following is a true statement regarding backup of the Cassandra datastore in the hybrid runtime plane?

A. Cassandra backup is enabled by default.

B. The Cassandra backup process does not need authentication to perform the backup.

C. Apigee hybrid can only back up Cassandra to a Cloud Storage bucket.

D. The Cassandra backup schedule is configured in the overrides configuration file using standard crontab syntax.

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Question #3

Answer

Which of the following is a true statement regarding backup of the Cassandra datastore in the hybrid runtime plane?

A. Cassandra backup is enabled by default.

B. The Cassandra backup process does not need authentication to perform the backup.

C. Apigee hybrid can only back up Cassandra to a Cloud Storage bucket.

D. The Cassandra backup schedule is configured in the overrides configuration file using standard crontab syntax.

Google Cloud

Explanation:

A. Incorrect. Cassandra backup is not enabled by default. It is a good practice to enable backups using the overrides configuration file.

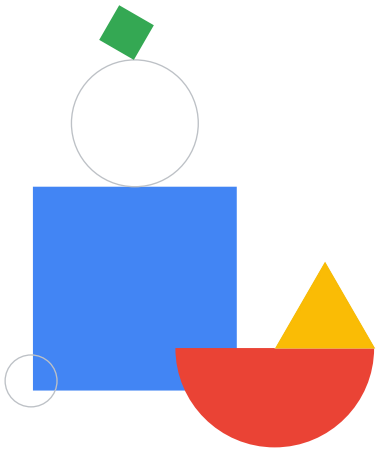
B. Incorrect. A service account with storage.objectAdmin role is required to perform the backup to Cloud Storage. Backups to the file system use SSH keys.

C. Incorrect. You can configure Cassandra backups to an external file system.

D. Correct! You configure the backup schedule using the schedule property in the overrides configuration file.

Google Cloud

Review: Installation and Platform Operation



In this module you learned how to install and configure Apigee hybrid on a Kubernetes platform. You installed hybrid on Google Kubernetes Engine in a lab.

You also learned how to use the helm tool to initialize and configure the hybrid runtime cluster.

Finally, you learned about the backup and restore process for the runtime datastore component used in hybrid.